Three sample chapters

100 Puzzles

Stories to Solve with Mathematics, Logic and Science

The award book of Math Club & Competitions.

This sample holds the Prologue, where the journey begins, and three puzzles from Volume 1: *The Gate of Four Keys* (heat and averages), *The Last Lotus Seed* (a winning strategy) and *Just Enough Salt* (balance and proof), each with its hint and full solution. Try each puzzle before you turn to the answers.

MATH CLUB & COMPETITIONS



PROLOGUE

The Silent Places

Snow had been falling on Highmere for three days when Master Orla called her four apprentices to the Hall of Maps. The Great Map filled the whole north wall: mountains in brown ink, rivers in blue, roads in fine red lines, a name beside every town. Copies of it hung in every inn and ferry house in the land. Travellers planned their journeys by it, healers found their way to the sick by it, and merchants sent their wagons along its roads.

But here and there a place was covered by a square slip of pale paper, pasted over the ink. On some slips Orla had written, in small brown letters, the last news that had come from there: *Gate shut since autumn. Lighthouse dark.* Others were blank. There were a hundred of them, and every year there were a few more.

"Every one of those places is still there," said Orla, tapping a slip with her cane. "A harbour, a market, a village on a cliff. People live there as they always have. But each of them has fallen silent. A gate that will not open, a light that will not burn, a quarrel nobody can settle: one trouble, and the wagons stop going, the letters stop coming, and the road grows over. A mapmaker draws only what she knows is true. When a place falls silent, I can no longer promise that its gate opens or its light burns, and a traveller who trusts my map could walk into danger. So I paste

a slip over it, here and on every copy, write down the last news I had, and wait. Where the slip is blank, nothing has been heard at all."

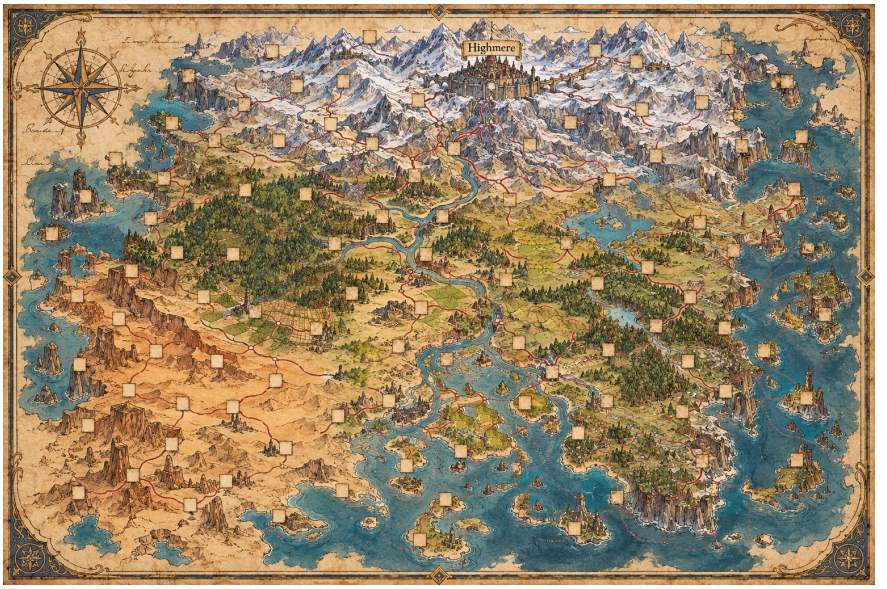
"Then we go and fix them," said Tavi.

"You go and *help*," said Orla. "Mend each trouble, truly and carefully, and people will travel there again. Then I can lift the slip, and add whatever new your sketch shows me: a path, a channel, a road."

She gave them until next midwinter, when they would meet again at the mountain gate. "Not all together: you will travel by three roads, and change companions as the seasons change. Write to me often, and I will write back with news of the others. Quill will carry the letters. When you help a place, put a sketch of it in your letter, and I will lift its slip." On the windowsill, a raven with one silver feather shook out her wings.

Orla's voice grew quieter. "One more thing. Long ago, another apprentice of mine set out to find these places, and never came home. If you ever find signs of someone who went before you, let me know as soon as possible."

That night the four of them set out under falling snow: three lanterns going three different ways down the mountain, and one raven who would fly between them and Highmere.



The Great Map of Highmere. The pale slips mark the hundred places that have fallen silent.





1 — The Gate of Four Keys

By the time Mira reached Coldhearth, a day out of Highmere, the sun had set and the path behind her was vanishing under snow. Ahead, a stone gate closed the narrow pass from cliff to cliff. Beyond it ran the road to the next valley, and no traveller had come down that road since autumn.

An old keeper stood by the gate with a lantern, his beard white with frost. "It will not open for me any more," he said. "Its four keys never leave it. They sleep in four drawers in the gate itself: two in the north wall, frozen into the ice, and two in the warm nook where the dragon sleeps."

In the nook, a little copper dragon named Ember lay curled so close to the warm stone that her scales glowed.

"The lock opens only when all four keys are at exactly the same temperature," said the keeper. "Metal grows a little when it is warm; a key even a tiny bit too warm or too cold will not turn."

Each drawer was marked with its key, and all four keys were made of the same metal, though not all of the same size.

Key	A	B	C	D
Mass	100 g	200 g	200 g	400 g
Temperature	0°C	90°C	90°C	0°C

In the middle of the gate sat a brass box with two hollows, one key in each, and each drawer had a brass handle. "Here is everything you may do," said the keeper.

1. Pull the handles of any two drawers. Those two keys slide into the box, and heat flows from the warmer key to the cooler one until both are equally warm. No heat escapes. Then both keys slide back into their own drawers.
2. Do it as often as you like, with any pair, the same key more than once if you wish. Each use of the box takes an hour.

"The drawers are lined with felt and sealed. While you work, a key in its drawer stays exactly as warm as it is."

Mira looked up at the sky. The storm would close the pass by morning, and once it closed, nobody could wade down through the snow.

WHAT YOU NEED TO KNOW

A heavier key needs more heat to warm up by one degree. Since all four keys are made of the same metal, two keys leave the box at the temperature

$$T = \frac{m_1 T_1 + m_2 T_2}{m_1 + m_2},$$

where m_1, m_2 are their masses and T_1, T_2 their temperatures before the box. For example, a 300 g key at 20°C and a 100 g key at 60°C both leave at

$$\frac{300 \cdot 20 + 100 \cdot 60}{300 + 100} = 30^\circ\text{C}.$$

All temperatures are exact.

CAN YOU HELP MIRA?

1. What temperature must all four keys reach? Why?

2. Find a plan that brings all four keys to that temperature. Give the pairs and their order.

3. What is the fewest uses of the box that can work? Explain why no shorter plan works.

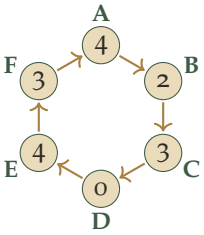




6 — The Last Lotus Seed

Reeds whispered along the water as Suri stepped off the ferry at Reedhaven, where the river splits into a hundred channels before the sea. Bao was still on the desert road, but Orla's last letter had brought his sketch of a wren disc from Sandglass, and Suri had copied it into her notebook.

Reedhaven's floating market has only one way in, and for two years Oru the water buffalo has kept it. Every boat must pay him a heavy toll. He has one custom: whoever beats him at a game of lotus seeds goes in free for the whole season. But nobody ever has, even though he always lets the visitor go first. The toll was too high, the traders went to other markets, and the market grew quiet.



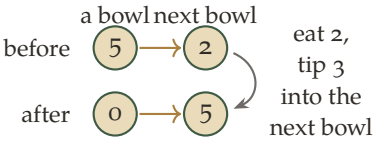
Under his striped awning at the entrance, Oru was polishing a round wooden tray, his wooden toll box beside it. Six bowls were fixed in a ring on it, marked A to F clockwise. A painted arrow ran from each bowl to the next: A to B, B to C, and so on, with F's arrow pointing back to A. Some bowls held fresh lotus seeds.

By the entrance an old woman with a boat of greens was counting out her toll coin by coin. Suri studied the tray, copied the six numbers into her notebook, and stepped up. "May I play for

her?" Boatmen gathered round, curious: a little girl in glasses, hoping to beat Oru! He chuckled. "You may. You go first. Whoever eats the last seed wins."

"Here is everything you may do," said Oru. "We take turns, and on your turn you:"

1. Choose a bowl that has seeds.
2. Eat one or two seeds from it (two only if it holds at least two), then tip all the seeds left in it into the next bowl along the arrow, even if that bowl is empty. That ends your turn.



"No passing, and no touching the seeds in any other way."

Out on the water a ferry bell rang, and the reeds went on whispering.

WHAT YOU NEED TO KNOW

Eaten seeds leave the tray; tipped seeds stay on it. All the seeds left go into *one* next bowl, joining any already there. The bowls never move, an empty bowl stays in the ring, and a bowl can hold any number of seeds.

CAN YOU HELP SURI?

1. Explain why a player with seeds on the tray is never stuck, and why every game must end.
2. Find a first move and a plan that let Suri eat the last seed however Oru plays. Prove that the plan always works.
3. Having lost, Oru demands a rematch: he adds one seed to bowl D, leaves the others as they were, and Suri goes first again. Can her first move still leave every bowl holding as many seeds as the bowl straight across from it? Explain.





9 — Just Enough Salt

The tide was coming in. Frost glittered on the salt grass as Tavi reached the place where the dike had been cut. Quill dropped onto Tavi's shoulder with a letter from Orla, which would have to wait.

Brinewick lies in the middle of a coastal salt marsh. The only way into the village is this narrow earth dike; on both sides of it are deep tidal channels, and in winter their water is icy and fast, so nobody can wade or swim across. Last autumn a storm sank the village's boats and dug a sinkhole right through the dike. Sea water rushes through the hole in a whirlpool. The same storm cracked the gear shaft of the sea sluice gate at the edge of the village, so the gate can no longer be lowered. Since then, every high tide has pushed salt water into the village, spoiling the fresh water well and soaking into the rice store.

Only one stone pillar of the old bridge is left in the middle of the hole. The villagers laid a plank across it, in a notch on top of the pillar. The plank's two ends hang just above the crumbling mud edges and do not rest on the banks. Nobody heavier than Fen can cross that plank, and nobody would send a child alone over the whirlpool, so the whole village is cut off, living on the last of its rice; no smoke rises from its kitchens.

On the far side, Fen, the old otter who keeps the sluice gate, waved wildly when he saw Tavi's tool belt. "Are you a mechanic? Tonight is the highest tide of the

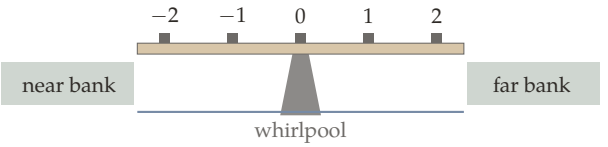
winter. If the sluice gate isn't mended before high water, the salt water will ruin the rest of the rice and the last well."

"Let me try!" Tavi put a foot straight onto the end of the plank. It dipped, the far end rose, and Fen shouted, "Take it off!"

"The plank only stays in its notch while the two sides nearly balance," he said, once Tavi had stepped back. "I tested it with salt: a 30 kg test sack on the end cleat, and it stays put; any heavier and the plank slips out of the notch. I weigh 20 kg, so I can cross alone, and for months I'm the only one who has brought a little rice into the village. And you?"

"Forty kilograms, tools and all," said Tavi. "And the tools stay on: the belt is too heavy for you, and I'll need every one at the gate." He looked at the plank again. The farther from the pillar you stand, the more easily you tip it.

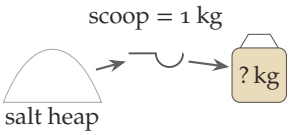
Five wooden cleats are nailed across the plank, one metre apart. The wet, frosty plank is too slippery to stand on anywhere else. Tavi numbered the cleats from −2 to 2; cleat 0 is right above the pillar.

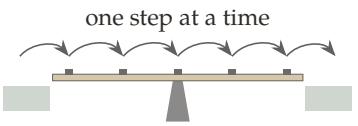


Beside Tavi stood the autumn's salt. The villagers had carted it out this far to wait for the traders, and then the sinkhole blocked the road. Now the waves were licking at the foot of the heap. Next to it lay an empty sack and a scoop; each scoop of salt weighs exactly 1 kg. The marsh is all mud, without a single stone.

"Here is everything we may do," said Fen.

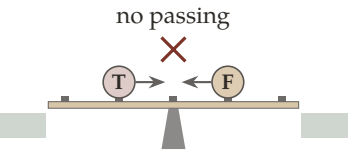
1. Before anyone steps onto the plank, Tavi puts as many scoops of salt into the sack as he likes. After that, nothing can be added or taken out: the waves will soon carry the whole heap away.

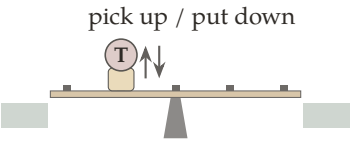




2. Only one of us moves at a time, and only one step: to the next cleat, or between an end cleat and the bank right next to it. The other stands still, so the plank does not rock.

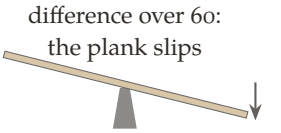
3. Two of us cannot stand on the same cleat or pass each other: the plank is only one foot wide.





4. Whoever stands somewhere may pick up the sack there, or put it down there, on a cleat or on a bank.

5. The plank must always be safe (see the box below). Otherwise it slips out of its notch and throws us into the hole.



Fen may end on either bank: alone, he can always cross back. Quill, who weighs almost nothing, flew off to watch from the reeds, as the water rose another notch.

WHAT YOU NEED TO KNOW

How strongly a weight tips the plank is called its *turning effect* (or *moment*). Here, turning effect = mass (kg) × metres from the weight to cleat 0. Add up the turning effects of everything on one side of cleat 0, then, separately, of everything on the other side. The plank is safe when the two totals differ by no more than 60, exactly the difference made by Fen's 30 kg test sack at cleat 2 ($30 \times 2 = 60$). Anything at cleat 0 or on a bank does not tip the plank; a carried sack is where its carrier is. For example, 30 kg at cleat -1 and 10 kg at cleat 2 give 30 on the near side and 20 on the far side, a difference of 10, so the plank is safe.

CAN YOU HELP TAVI AND FEN?

1. Why, with no salt at all, can Tavi never reach the far bank?
2. How many scoops of salt should Tavi put in the sack? Find a safe way for him to cross.
3. Prove that with any other amount of salt, Tavi cannot cross.

BEFORE YOU READ ON

Have you tried every question? The hints come first, then the full solutions.

1 The Gate of Four Keys *page 4*

First ask: which use of the box can be the *first* to make a key reach 40°C?

6 The Last Lotus Seed *page 6*

After Suri's first move, compare each bowl with the bowl straight across the tray from it.

9 Just Enough Salt *page 8*

Think about the first moment Tavi stands at cleat 2. Something must then be on the near side of cleat 0. What, and how did it get there?

Still stuck? At some point the sack must be carried from the near bank onto cleat -2. Who can carry it there, and who can be on the plank at that moment?

SOLUTION 1

The Gate of Four Keys

Tags: Weighted averages; Heat transfer; Conservation; Logical deduction

1. The only possible common temperature

For each key, multiply its mass by its temperature, then add the four results. This sum never changes: when two keys leave the box at temperature T, the rule gives $(m_1 + m_2)T = m_1T_1 + m_2T_2$, and the other two keys are untouched. At the start the sum is $100 \cdot 0 + 200 \cdot 90 + 200 \cdot 90 + 400 \cdot 0 = 36\,000$. If all four keys end at the same temperature t, then

$$(100 + 200 + 200 + 400)\, t = 36\,000, \qquad \text{so} \qquad t = 40.$$

All four keys must reach $t = 40°\text{C}$.

2. A plan with four uses

One plan that works is:

		Temperature (°C)			
Use	**Pair**	A	B	C	D
0	Start	0	90	90	0
1	B and D	0	30	90	30
2	A and C	60	30	60	30
3	C and D	60	30	40	40
4	A and B	40	40	40	40

The calculations are

$$\text{Use 1: } \frac{200 \cdot 90 + 400 \cdot 0}{200 + 400} = 30, \qquad \text{Use 2: } \frac{100 \cdot 0 + 200 \cdot 90}{100 + 200} = 60,$$

$$\text{Use 3: } \frac{200 \cdot 60 + 400 \cdot 30}{200 + 400} = 40, \qquad \text{Use 4: } \frac{100 \cdot 60 + 200 \cdot 30}{100 + 200} = 40.$$

Any valid four-use plan is accepted, for example (A, B), (C, D), (A, C), (B, D).

3. Why fewer uses cannot work

The first use cannot make any key reach 40°C. Checking the starting pairs gives:

Starting pair	Temperature after the box
A and D	0°C
B and C	90°C
A and B, or A and C	60°C
B and D, or C and D	30°C

So two uses cannot be enough: after use 1 no key is at 40°C, and use 2 changes only two keys.

Now suppose three uses are enough. Use 3 changes only two keys, so the other two are already at 40°C after use 2; since none was after use 1, they are exactly the two keys of use 2. The remaining two keys must both enter the box in use 3, and they have not changed since use 1. So right after use 1, the keys must split into two pairs such that each pair, put in the box together, comes out at exactly 40°C. We show that no first use allows this. (Remember that two keys always come out at a temperature between their own.)

Case 1. The first use pairs keys at the same temperature. Their temperatures do not change, so the starting arrangement remains unchanged. The table above shows that no pair comes out of the box at 40°C.

Case 2. The first use produces two keys at 60°C. The four temperatures are now 60, 60, 90, 0. For the 90°C key and its partner to come out at 40°C, the partner must be colder than 40°C. The only one is the 0°C key. That leaves the two 60°C keys together, and they come out at 60°C, not 40°C.

Case 3. The first use produces two keys at 30°C. The four temperatures are now 30, 30, 90, 0. For the 0°C key and its partner to come out at 40°C, the partner must be warmer than 40°C. The only one is the 90°C key. That leaves the two 30°C keys together, and they come out at 30°C, not 40°C.

Every possible first use has been covered, and none allows a three-use solution. Since four uses do work,

The fewest is **4 uses**: four hours at the gate.

On the map *The gate swung open before dawn, and the keeper walked Mira to the first bend of the road up to the pass. Ember said goodbye to the keeper and trotted after her. That night Mira drew Coldhearth, and Quill took the sketch home to Orla.*

SOLUTION 6

The Last Lotus Seed

Tags: Strategy game; Symmetry; Logical deduction

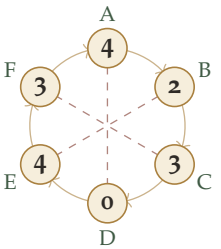
1. Never stuck, and always an end

If any seeds remain, a player can choose a bowl that has some and eat one. Tipping only moves seeds, so every turn the tray loses exactly 1 or 2 seeds. It starts with $4+2+3+0+4+3=16$, so the game ends after at most 16 turns, when the tray is empty.

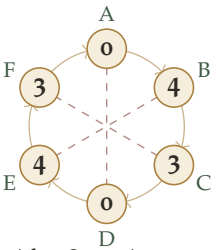
> Any bowl with seeds allows a move, and the total falls every turn, so the game must end.

2. Suri's first move: make the two halves match

Across the tray, A faces D, B faces E, and C faces F. Suri chooses A and eats 2; the other 2 seeds go into B.



Start



After Suri: A, eats 2

Now every bowl holds as many seeds as the bowl straight across from it (the dashed lines join facing bowls).

3. Then copy Oru across the tray

From now on, whenever Oru chooses a bowl and eats some seeds, Suri chooses the bowl straight across from it and eats the same number.

Why this is always allowed: before Oru moves, say his bowl holds x seeds and the next bowl holds y. The two bowls across from them also hold x and y. Oru's move changes only his two bowls: his becomes empty and the next one gets $y + x - k$, where k is the 1 or 2 he ate. On a ring of six, two neighbouring bowls and the two bowls across from them are four *different* bowls, so Oru has not touched Suri's two. Her bowl still holds $x \geq k$ seeds, so she may eat k.

Why the halves match again: her move empties her bowl and gives the next one $y + x - k$ seeds, exactly like Oru's pair. The third pair of facing bowls was not touched at all. This works whichever bowl Oru picks, for $k = 1$ or 2, even when $x = k$ (nothing is tipped) or $y = 0$. (Copying across keeps the arrows' direction: A→B faces D→E.)

4. Why Suri eats the last seed

After any move by Oru, the bowl across from his still holds $x \geq 1$ seeds, so Oru can never empty the tray. The game must end (part 1), so it ends on Suri's turn.

> Choose A and eat 2. Then always answer Oru at the bowl straight across from his, eating the same number. Suri eats the last seed. Another first move counts only with a proof that it always wins.

One possible game with Suri's plan:

Player	Move	A	B	C	D	E	F
Suri	A, eats 2	0	4	3	0	4	3
Oru	C, eats 1	0	4	0	2	4	3
Suri	F, eats 1	2	4	0	2	4	0
Oru	B, eats 2	2	0	2	2	4	0
Suri	E, eats 2	2	0	2	2	0	2
Oru	A, eats 2	0	0	2	2	0	2
Suri	D, eats 2	0	0	2	0	0	2
Oru	C, eats 2	0	0	0	0	0	2
Suri	F, eats 2	0	0	0	0	0	0

5. The rematch

Now the tray is $4, 2, 3, 1, 4, 3$, and two facing pairs are unequal: A and D, and B and E. A move changes only the chosen bowl and the next one, so to fix both pairs Suri must choose A (changing A and B) or D (changing D and E). But choosing A empties A while D still holds 1, and choosing D empties D while A still holds 4.

> No. No first move makes the halves match, so the copying plan cannot start.

On the map *Suri won, then showed the whole quay how. Oru laughed (he had been beaten fairly, and a custom is a custom), lowered the toll to what it used to be, and the traders came back. He poled Suri round the channels to draw them for Orla.*



SOLUTION 9

Just Enough Salt

Tags: Turning effect (moments); Logical deduction; Proof by cases

1. The two most dangerous moments

Tavi can only step onto the plank at cleat -2 (from the near bank), and can only step off onto the far bank from cleat 2. Look at what the plank needs at those two moments.

- When Tavi stands at cleat -2, he gives the near side $40 \times 2 = 80$. For the difference to be no more than 60, the far side must have at least $80 - 60 = 20$.
- When Tavi stands at cleat 2, he gives the far side 80. In the same way, the near side must have at least 20.

Apart from Tavi, only Fen and the sack can be on the plank. Fen weighs 20 kg, so he gives 20 when he stands at cleat 1 or -1, and 40 at cleat 2 or -2.

One more thing: on the plank nobody can pass anybody. So as long as Tavi stays on the plank, Fen cannot get from in front of him (the far-bank side) to behind him (the near-bank side), or the other way round.

2. No salt: Tavi is stuck

With no salt, only Fen can supply that "at least 20".

When Tavi steps onto cleat -2, the far side needs at least 20, so Fen must be standing at cleat 1 or 2: *in front of* Tavi. From then on, as long as Tavi stays on the plank, Fen stays in front of him. When Tavi reaches cleat 2, there is no cleat in front of him, so Fen can only have stepped off onto the far bank. The near side is then empty: 0, while the far side has 80. A difference of 80 is more than 60: the plank slips.

If Tavi goes back to the near bank and tries again, exactly the same thing happens.

> With no salt, Tavi can never stand safely at cleat 2, so he can never reach the far bank.

3. Ten scoops are enough

Put in 10 scoops (10 kg). The table shows each step, where everything is after it, and the turning effect on each side.

	Step	Tavi	Fen	Sack	Near	Far	Diff.
1	Fen steps onto cleat 2	bank	2	bank	0	40	40
2	Tavi picks up the sack	bank	2	Tavi	0	40	40
3	Tavi carries it to cleat -2	-2	2	Tavi	100	40	60
4	Tavi puts it down	-2	2	-2	100	40	60
5	Tavi to cleat -1	-1	2	-2	60	40	20
6	Tavi to cleat 0	0	2	-2	20	40	20
7	Tavi to cleat 1	1	2	-2	20	80	60
8	Fen steps onto the far bank	1	bank	-2	20	40	20
9	Tavi to cleat 2	2	bank	-2	20	80	60
10	Tavi steps onto the far bank	bank	bank	-2	20	0	20

The difference is never more than 60. The sack stays at cleat -2; Fen can still cross alone safely afterwards.

4. Less than 10 kg is not enough

Look at the first time Tavi stands at cleat 2, and at the last time before that when he stepped from the near bank onto cleat -2. In between, he stayed on the plank. There are two cases, depending on where Fen was when Tavi stepped onto cleat -2.

Fen was on the plank. Then Fen was in front of Tavi, and stays in front until Tavi reaches cleat 2. By then Fen has stepped off onto the far bank, so he adds nothing to the near side. So the near side's "at least 20" must come from the sack alone.

Fen was on a bank. Then, just as Tavi stepped onto cleat -2, Fen added nothing to the far side. So the far side's "at least 20" must come from the sack alone.

Either way, at some moment the sack alone gives a turning effect of at least 20. The sack is never more than 2 metres from cleat 0, so it weighs at least $20 \div 2 = 10$ kg.

5. More than 10 kg is too much

By part 4, the sack must be on the plank at some moment. It starts on the near bank, so someone must carry it from the near bank onto cleat -2. Look at the first time that happens. Call the sack's mass w kg.

Tavi carries it up. The near side has $(40 + w) \times 2 = 80 + 2w$. The far side has at most 40, with Fen at cleat 2. To be safe we need $80 + 2w \leq 40 + 60 = 100$, that is, $w \leq 10$.

Fen carries it up. He has just stepped up from the near bank. Until now the sack has never been on the plank, so every time Tavi was on the plank, Fen was in front of him (as in part 2). But now Fen is on the near bank, which would be behind Tavi if

Tavi were on the plank; so Tavi cannot be on the plank. Nor can Tavi have reached the far bank yet, since for that the sack must be on the plank first (part 4). So Tavi is on the near bank, and the far side is empty. The near side has $(20 + w) \times 2$, so we need $40 + 2w \leq 60$, that is, $w \leq 10$.

> Tavi must put in **exactly 10 scoops**, 10 kg of salt. Any less and the plank slips when Tavi reaches the far end; any more and it slips as the sack is carried up.

On the map *That night Tavi clamped the cracked gear shaft with iron bands, and the sluice gate came down just before high water. Fen showed him which banks of the marsh stay dry at the highest tides. Tavi sent his drawing with Quill, the paper still dusted with salt.*